



Gosford High School

Year 11

2009

Preliminary Higher School Certificate

Mathematics

Assessment Task 4

Time Allowed – 2 hours

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

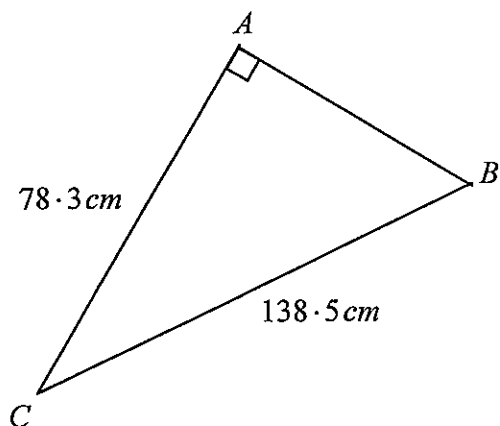
Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

QUESTION 1 (13 marks)

(a)



Find the length of the interval AB correct to 3 significant figures.

(2)

- (b) A bicycle is bought for \$840 and sold 3 years later for \$315.
Express the loss as a percentage of the cost price.

(1)

- (c) Expand and simplify $(x-3)^2 + 5x$

(2)

- (d) Solve $\frac{3x}{2} + 4 = x$

(1)

- (e) Simplify $\frac{3x^2 - 5x - 2}{9x^2 - 1}$

(2)

- (f) Find the value of x and y if $x + \sqrt{y} = \frac{3}{2\sqrt{3} - 3}$

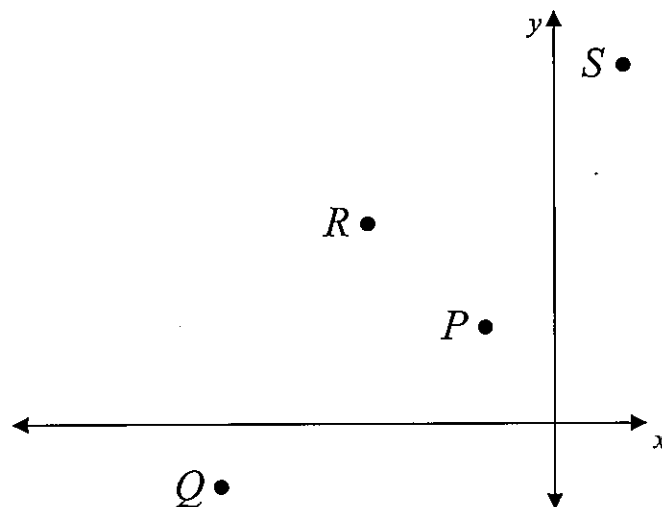
(2)

- (g) Solve $|2x| = 5x + 9$

(3)

QUESTION 2 (13 marks)

- (a) A quadrilateral has vertices $P(-1\frac{1}{2}, 2)$, $Q(-6, -1)$, $R(-3\frac{1}{2}, 4)$ and $S(1, 7)$.



- (i) Find the midpoint of PR (1)
 - (ii) Show that PQRS is a parallelogram (2)
 - (iii) Find the length of QS (1)
 - (iv) Find the gradient of QS (1)
 - (v) Find the equation of QS in general form (2)
 - (vi) Find the perpendicular distance from P to QS (2)
 - (vii) Find the area of the parallelogram PQRS (2)
- (b) Find the point of intersection of the lines $x + y + 3 = 0$ and $3x - 4y + 2 = 0$ (2)

QUESTION 3 (24 marks)

- (a) Sketch $y = 2^{-x}$ neatly on a number plane. (2)
- (b) Sketch $y = x^2 + 2x - 3$ neatly, showing the coordinates of the vertex and the intercepts with the coordinate axes. (3)
- (c) Shade the region satisfying $y < |x - 1|$ and $x \geq 0$ (3)
- (d) Sketch the curve $y = \sqrt{4 - x^2}$ (2)
- (e) Given $F(x) = \sqrt{2 - x}$
- (i) Find $F(-3)$ (1)
- (ii) Find x if $F(x) = 0$ (1)
- (iii) State the domain and range of the function $y = F(x)$ (2)
- (iv) Sketch the curve $y = F(x)$ (2)
- (v) Find the point of intersection of the curve $y = F(x)$ and the line $x + y = 0$ (3)
- (f) Consider the function $g(x) = \frac{\cos x}{x}$
- (i) Show that the function is an ODD function (2)
- (ii) For what value of x is the function $y = \frac{\cos x}{x}$ undefined (1)
- (iii) Find where the curve $y = \frac{\cos x}{x}$ crosses the x axis, if $0 \leq x \leq 360^\circ$ (2)

QUESTION 4 (13 marks)

- (a) Write down the exact value of $\sec(135)^\circ$ (1)
- (b) A triangle has sides 7cm, 4cm and 5cm.
Find the size of the largest angle in the triangle correct to the nearest minute. (2)
- (c) Simplify $\frac{\sqrt{1 - \sin^2 A}}{\cot A}$ (2)
- (d) If $\operatorname{cosec} \theta = -2$ and $\cos \theta > 0$, find the exact value of $\tan \theta$. (2)
- (e) Solve $2 \cos^2 x + \cos x = 0$ for $0 \leq x \leq 360^\circ$ (3)
- (f) A fishing boat anchored at a point A observes a lighthouse L on a bearing of 050°N .
The boat sails due North until it reaches and anchors at a buoy B.
Charts tell the fisherman that the lighthouse L is 30 km from buoy B and has a bearing of 125°N .
- (i) Draw a neat diagram, showing clearly on your diagram all the relevant given information. (1)
- (ii) Find the distance from point A to the buoy B. (2)

QUESTION 5 (24 marks)

- (a) Find the solutions to equation $2x^2 - 4x - 1 = 0$ in simplest surd form (3)
- (b) Solve $x(3 - x) \leq 0$ (1)
- (c) Find the minimum value of the expression $x^2 + 4x + 3$ (2)
- (d) If α and β are the roots of the equation $x^2 - 2x + 5 = 0$, find
- (i) $\alpha + \beta$ (1)
 - (ii) $\alpha\beta$ (1)
 - (iii) $(\alpha - 2)(\beta - 2)$ (2)
 - (iv) $\alpha^2 + \beta^2$ (2)
- (e) Solve $(x^2 - 2x)^2 - (x^2 - 2x) - 6 = 0$ (3)
- (f) Find A , B and C if $x^2 - 3x + 6 \equiv A(x - 2)^2 + Bx + C$ (3)
- (g) Find the range of values of k for which the equation $2x^2 - 3kx + 2 = 0$ has no real roots (2)
- (h) Find the values of m if the equation $x^2 - 3mx + (m + 3) = 0$ has
- (i) one root is the reciprocal of the other (1)
 - (ii) roots that are equal but opposite in sign (1)
 - (iii) one root is double the other (3)

QUESTION 6 (13 marks)

(a) Find $\frac{d}{dx}[(2x-1)^2]$ (2)

(b) Find $\frac{dy}{dx}$ if $y = 4x^3 - \frac{3}{x} + 5$ (2)

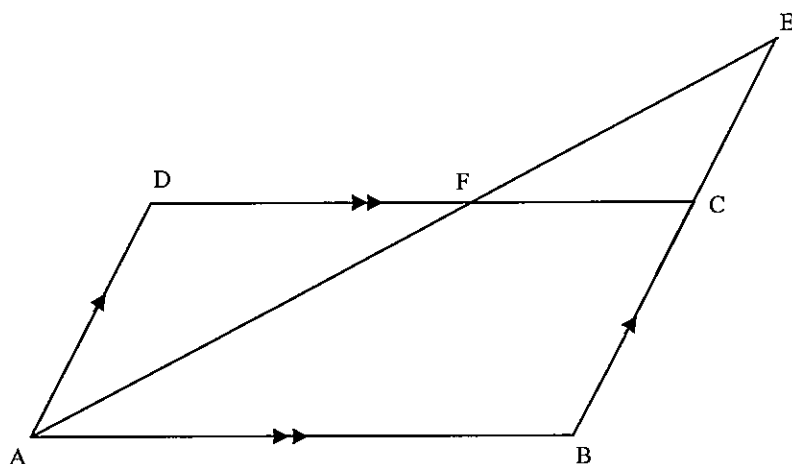
(c) Find $f'(4)$ if $f(x) = 2\sqrt{x}$ (2)

(d) Use first principles to find the gradient of the tangent to the curve $y = x^2 - 3x$ (2)

(e) In the diagram below ABCD is a parallelogram.
F lies on DC such that AF produced meets BC produced at E

(i) Prove $\triangle ADF \parallel \triangle EBA$ (3)

(ii) Find DF if $FC = 6\text{cm}$, $EC = 5\text{cm}$ and $CB = 8\text{cm}$ (2)



Solutions to 2009 Preliminary HSC Maths Task 4

Q1 a) $c^2 = 138.5^2 - 78.3^2$
 $= 13051.36$
 $c \doteq 114 \text{ cm} \quad (3 \text{ sig figs})$

b) Loss = \$ 525
 $\% \text{ loss} = \frac{525}{840} \times 100 \%$
 $= 62.5 \%$

c) $x^2 - 6x + 9 + 5x = x^2 - x + 9$

d) $3x + 8 = 2x$
 $x = -8$

e) $\frac{(3x+1)(x-2)}{(3x+1)(3x-1)} = \frac{x-2}{3x-1}$

$\begin{array}{r} 3x + 1 \\ \times \quad x - 2 \\ \hline \end{array}$

f) $\frac{3}{2\sqrt{3}-3} \times \frac{2\sqrt{3}+3}{2\sqrt{3}+3} = \frac{6\sqrt{3}+9}{12-9}$
 $= \frac{3(2\sqrt{3}+3)}{3} = 2\sqrt{3}+3$

g) $|2x| = 5x + 9$

$2x = 5x + 9$

$-9 = 3x$

$-3 = x$

or $-2x = 5x + 9$

$-9 = 7x$

$-\frac{9}{7} = x$

Check $|-6| \stackrel{?}{=} -15 + 9$
 No

$|\frac{-18}{7}| \stackrel{?}{=} -\frac{45}{7} + 9$
 Yes

$\therefore x \doteq -\frac{9}{7} \text{ only}$

Q2

$$i) \left(-\frac{5}{2}, \frac{6}{2}\right) = (-2\frac{1}{2}, 3)$$

$$ii) \text{Midpoint of } QS: \left(\frac{-6+4}{2}, \frac{-1+7}{2}\right) \\ = (-2\frac{1}{2}, 3)$$

Equal midpoints $\therefore$ Diagonals bisect each other. $\therefore$ PQRS is a parallelogram.

$$iii) d^2 = 7^2 + 8^2 \\ d = \sqrt{113}$$

$$iv) m = \frac{7+1}{1+6} = \frac{8}{7}$$

$$v) y-7 = \frac{8}{7}(x-1)$$

$$7y - 49 = 8x - 8$$

$$0 = 8x - 7y + 41$$

$$vi) d = \left| \frac{8x - \frac{7}{2} - 7 \times 2 + 41}{\sqrt{8^2 + 7^2}} \right| \quad \left(-1\frac{1}{2}, 2\right)$$

$$d = \left| \frac{15}{\sqrt{113}} \right| = \frac{15}{\sqrt{113}}$$

$$vii) A = bh = \sqrt{113} \times \frac{15}{\sqrt{113}} \\ = 15 \text{ m}^2$$

$$b) \quad x + y = -3 \quad \text{--- ①}$$

$$3x - 4y = -2 \quad \text{--- ②}$$

Sub into ①

$$-2 + y = -3$$

$$y = -1$$

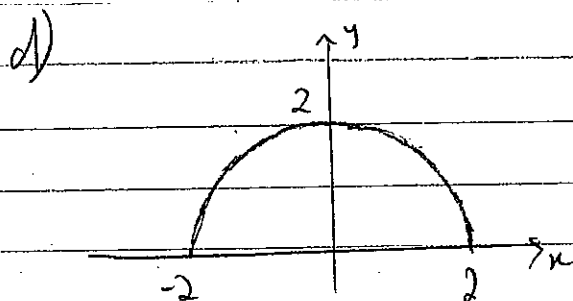
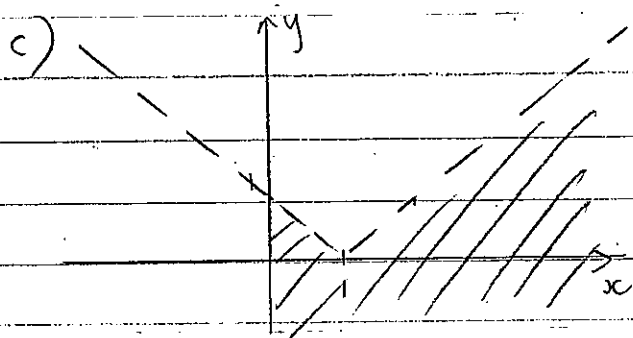
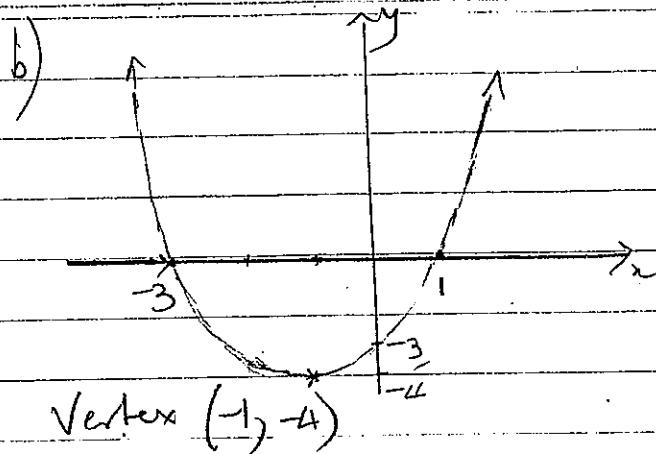
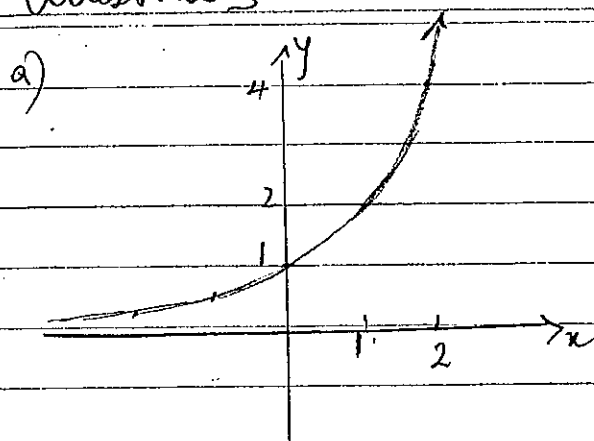
$$\textcircled{1} \times 4 \quad 4x + 4y = -12 \quad \text{--- ③}$$

$$\textcircled{2} + \textcircled{3} \quad 7x = -14$$

$$x = -2$$

$$(-2, -1)$$

Question 3



e)

$$i) F(-3) = \sqrt{5}$$

$$ii) 0 = \sqrt{2-x}$$

$$0 = 2-x$$

$$x = 2$$

iii)

$$D: x \leq 2$$

$$R: y \geq 0$$

v)

$$-x = \sqrt{2-x}$$

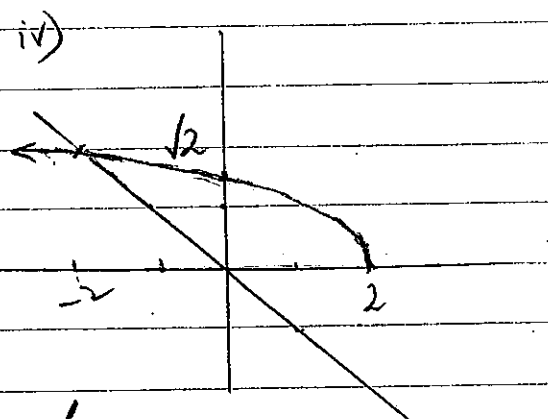
$$x^2 = 2-x$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \text{ or } 1$$

Only $x = -2$ works
Point of intersection is $(-2, 2)$



f)

$$g(x) = \frac{\cos x}{x}$$

$$g(-x) = \frac{\cos(-x)}{-x}$$

$$= \frac{\cos x}{-x}$$

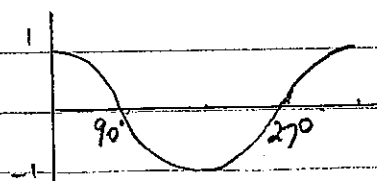
$$g(-x) = -g(x)$$

$\therefore$ odd.

iii)

$$\frac{\cos x}{x} = 0$$

$$\cos x = 0$$



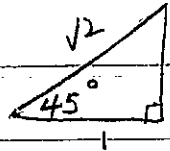
ii)

$$x = 0$$

$$x = 90^\circ \text{ or } 270^\circ$$

Question 4

a) $\sec 135^\circ = -\sec 45^\circ$
 $= -\sqrt{2}$



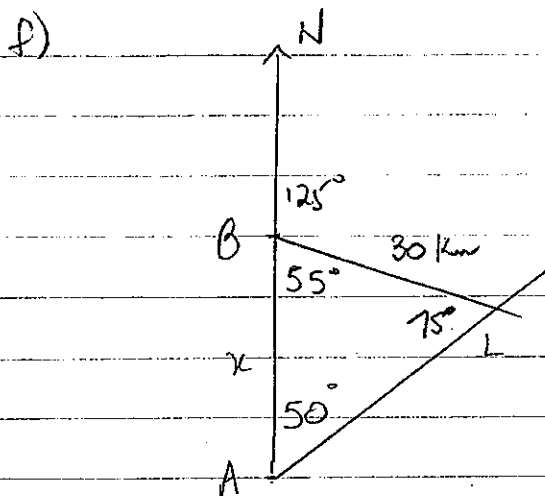
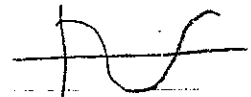
b) $\cos \theta = \frac{4^2 + 5^2 - 7^2}{2 \times 4 \times 5} = -\frac{1}{5}$
 $\angle \theta = 101^\circ 32'$



c) $\sqrt{\cos^2 A} \times \frac{\sin A}{\cos A} = |\cos A| \times \frac{\sin A}{\cos A}$
 $= \sin A$ if $\cos A > 0$ $= -\sin A$ if $\cos A < 0$
 ie Quads (1)+(4) ie quads (2)+(3)

d) $\sin \theta = -\frac{1}{2}$ θ lies in 4th quad
 $\theta = 330^\circ$ $\therefore \tan \theta = -\frac{1}{\sqrt{3}}$

e) $\cos x (2 \cos x + 1) = 0$
 $\cos x = -\frac{1}{2}$ or $\cos x = 0$
 $x = 120^\circ$ or 240° or 90° or 270°



ii) $\frac{x}{\sin 75^\circ} = \frac{30}{\sin 50^\circ}$
 $x = \frac{30 \sin 75^\circ}{\sin 50^\circ}$

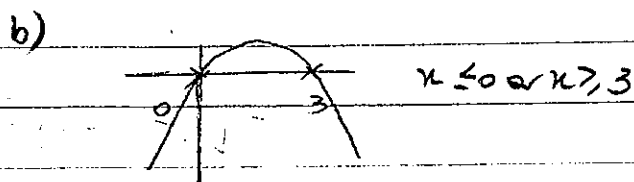
$x = 37.827798$

$x \approx 37.83 \text{ km}$

Question 5

$$a) \quad x = \frac{4 \pm \sqrt{16+8}}{8}$$

$$x = \frac{4 \pm 2\sqrt{6}}{8} = \frac{2 \pm \sqrt{6}}{4}$$



c)

$$x^2 + 4x + 4 - 1$$

$$(x+2)^2 - 1$$

Minimum Value is -1

d) i) $\alpha + \beta = 2$

ii) $2\beta = 5$

iii) $2\beta - 2(\alpha + \beta) + 4$
 $= 5 - 4 + 4$
 $= 5$

iv) $(\alpha + \beta)^2 - 2\alpha\beta$
 $= 4 - 10$
 $= -6$

e) $u = x^2 - 2x$

$$u^2 - u - 6 = 0$$

$$(u-3)(u+2) = 0$$

$$u = 3 \text{ or } -2$$

$$x^2 - 2x - 3 = 0 \text{ or } x^2 - 2x + 2 = 0$$

$$(x-3)(x+1) = 0 \quad d = 4 - 8$$

$$x = 3 \text{ or } -1$$

$= -4$
no solutions

$$\therefore x = 3 \text{ or } -1$$

f) Equate coeffs of x^2 $A = 1$

Sub $x = 0$ $4A + C = 6$

$$C = 2$$

Sub $x = 2$ $4 - 6 + 6 = 2B + 2$

$$2B = 2 \Rightarrow B = 1$$

$$A = 1, B = 1, C = 2$$

g) $2x^2 - 3kx + 2 = 0$

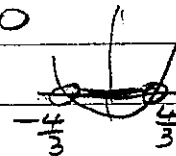
$$\Delta < 0$$

$$9k^2 - 4 \times 2 \times 2 < 0$$

$$9k^2 - 16 < 0$$

$$(3k-4)(3k+4) < 0$$

$$-\frac{4}{3} < k < \frac{4}{3}$$



h) i) $\alpha\beta = 1$

$$m+3 = 1$$

$$m = -2$$

ii) $b = 0$

$$-3m = 0$$

$$m = 0$$

iii) let roots be α & 2α

$$\therefore 3\alpha = 3m$$

$$\alpha = m$$

$$2\alpha^2 = m+3$$

$$2m^2 = m+3$$

$$2m^2 - m - 3 = 0$$

$$(2m-3)(m+1) = 0$$

$$m = \frac{3}{2} \text{ or } -1$$

$$\frac{2m-3}{m+1}$$

Question 6

a) $y' = 2(2x-1) \cdot 2 = 4(2x-1)$ or $8x-4$

b) $y = 4x^3 - 3x^{-1} + 5$

$y' = 12x^2 + 3x^{-2} = 12x^2 + \frac{3}{x^2}$

c) $f(x) = 2x^{\frac{1}{2}}$ $f'(x) = x^{\frac{1}{2}-1} = -\frac{1}{\sqrt{x}}$

$f'(4) = -\frac{1}{\sqrt{4}} = -\frac{1}{2}$

d) $y + \Delta y = (x + \Delta x)^2 - 3(x + \Delta x)$

$y + \Delta y = x^2 + 2x\Delta x + (\Delta x)^2 - 3x - 3\Delta x$

$\Delta y = 2x\Delta x + (\Delta x)^2 - 3\Delta x$

$\frac{\Delta y}{\Delta x} = 2x + \Delta x - 3$

$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = 2x - 3$

e) i) In Δ 's ADF , EBA

1. $\angle DAF = \angle FEC$ (alternate $\angle$'s $DA \parallel EB$)

2. $\angle ADF = \angle ABE$ (opposite angles of parallelogram are equal)

$\therefore \triangle ADF \parallel \triangle EBA$ (2 angles test)

ii) $\frac{DF}{AB} = \frac{AD}{EB}$ (corresponding sides in similar Δ 's)

$\frac{x}{x+6} = \frac{8}{13}$

$13x = 8x + 48$

$5x = 48$

$x = \frac{48}{5}$

$\therefore DF = \frac{48}{5}$ cm.

